

IIT Hyderabad

AI Based Recognition of Hand Drawn Chemical Structures

A Controlled Study of Synthetic Augmentation and Real Image Generalization

Mini Project Thesis

BT5230 AI in Drug Discovery

Challenge 72: Optical Chemical Structure Recognition for Digitizing Legacy Chemistry Literature

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Domain: AI in Drug Discovery, Cheminformatics, Computer Vision

Introduction and Project Objective

Chemical structures remain common in laboratory notebooks, teaching material, patents, and historical literature as drawings rather than machine-readable records. Optical chemical structure recognition (OCSR) attempts to recover a molecular representation from such an image. A successful system can reduce manual transcription and make structures available for search, descriptor calculation, similarity analysis, and downstream modelling. Hand-drawn images are especially difficult because stroke shape, bond geometry, lettering, contrast, background, and stereochemical marks vary between writers.

The project asked a narrow experimental question: does chemistry-aware, hand-drawn-style augmentation improve a fixed model’s generalization to real drawings when training data and compute are limited? The intended system accepts a cropped image containing one molecule and returns a SMILES string. The study does not address page-level detection, reaction schemes, clinical use, autonomous chemical database ingestion, or discovery of new drug candidates.

Problem Statement

Develop and fine-tune a deep-learning model that converts a cropped image containing one hand-drawn chemical structure into a valid canonical SMILES representation, then determine whether matched synthetic hand-drawn augmentation improves recognition on unseen real drawing styles.

Project Contribution

- A molecule-level, scaffold-separated ChEMBL dataset with matched clean and augmented depictions.
- A controlled MolScribe transfer-learning comparison in which only the training depiction condition changed.
- A held-out evaluation on 5,088 real hand-drawn images with invalid predictions retained in the denominator.
- Paired confidence intervals, chemical subgroup analysis, exact-match metrics, and fingerprint similarity.
- Runnable verification code and saved per-image outputs that reproduce the reported result without a GPU.

Reference Research and Technical Context

Primary Reference Paper

The primary reference is Rajan et al., *Advancements in hand-drawn chemical structure recognition through an enhanced DECIMER architecture*, published in the *Journal of Cheminformatics* in 2024 [1]. The paper uses an EfficientNetV2-M image encoder and a six-layer Transformer decoder to generate SMILES from 512×512 pixel images. Its models were trained for 25 epochs on TPU V4 hardware using millions of ChEMBL- and PubChem-derived depictions, including synthetic hand-drawn-like images from RanDepict. On the same 5,088-image real benchmark used here, the paper reports 73.25% identical predictions and average Tanimoto similarity of 0.94 for its best large-scale model.

That study motivates both the project hypothesis and the experimental design. It suggests that synthetic style diversity can help hand-drawn recognition, but its scale cannot be reproduced in a short course project. The present work therefore tests a smaller, controlled transfer-learning intervention rather than claiming a reproduction of DECIMER training.

Selected Transfer Model

MolScribe, introduced by Qian et al. [2], is an image-to-graph OCSR model that combines a Swin Transformer image encoder with Transformer-based decoding. The official pretrained checkpoint provided a viable starting point after compact scratch-trained models showed low validation performance. For this project, the supervised output format used `chartok_coords`, which predicts the tokenized SMILES sequence together with atom coordinates. The project dataset did not include the optional bond-edge targets.

Relationship to Drug Discovery

The model does not predict activity, toxicity, or binding. Its drug-discovery relevance lies upstream in data capture. Converting a drawing to a molecular graph allows the structure to enter cheminformatics workflows for similarity search, descriptor calculation, virtual screening, or archival curation. Because an incorrect structure can propagate silently into those workflows, exact connectivity, atom identity, charge, and stereochemistry matter.

Research Questions and Experimental Design

Table 1: Research questions and operational tests

Question	Operational test
RQ1	Measure clean fine-tuning on the held-out real hand-drawn benchmark.
RQ2	Compare matched augmented and clean checkpoints per image.
RQ3	Stratify outcomes by atom count, ring count, stereocenters, and SMILES length.
RQ4	Compare exact isomeric, exact non-isomeric, validity, and Tanimoto conclusions.

The main hypothesis was that augmented fine-tuning would improve exact recognition and mean molecular similarity over clean fine-tuning. A second hypothesis expected the intervention to help with visual variation while leaving stereochemistry as a major recognition challenge. Both hypotheses were evaluated as empirical claims rather than assumptions.

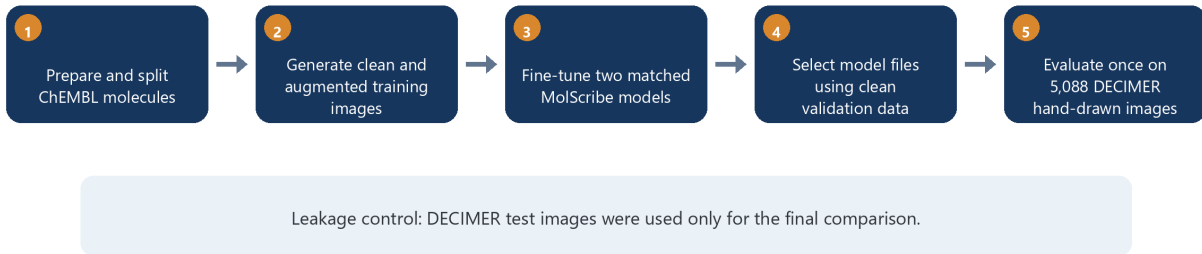


Figure 1: Controlled five-stage workflow. DECIMER test images were reserved for the final comparison and were not used to choose training or model settings.

Controls Against Leakage

- Molecules were split before rendering, and different depictions of one molecule could not cross partitions.
- Clean and augmented conditions used identical molecule IDs, labels, training count, validation images, starting checkpoint, seed, optimizer settings, and training duration.
- DECIMER test images were not used for training, validation, checkpoint selection, or augmentation design.
- All 5,088 benchmark rows remained in the denominator, including invalid model outputs and two reference SMILES unsupported by the local RDKit build.
- No model or augmentation setting was changed after the DECIMER results were calculated.

Data Preparation and Integrity Controls

Synthetic Molecule Set

The source was a distributed sample from ChEMBL 37 [7]. After canonicalization and filtering, 4,698 molecules were retained. The pipeline rejected 142 multi-fragment records and 160 records exceeding the selected sequence-length limit. A deterministic scaffold-aware split assigned 3,758 molecules to training, 470 to validation, and 470 to synthetic testing. No canonical molecule or scaffold crossed the partition boundary.

Table 2: Experimental data partitions

Partition	Molecules	Use
Training	3,758	Matched clean or augmented fine-tuning
Validation	470	Clean images for model viability and checkpoint assessment
Synthetic test	470	Held out and excluded from Colab training packages
Real benchmark	5,088	Final external evaluation only

The authoritative input to the experimental pipeline was `data/manifests/chembl37_stage_5000.csv`, in which the `split` column records the assignment of every molecule. This master manifest was used to construct the condition-specific image manifests and the Colab training packages used for the reported experiments. For direct evaluator inspection and compliance with the requested submission layout, the same unchanged rows are additionally supplied as `splits/train.csv`, `splits/validation.csv`, and `splits/test.csv`. These three files are convenience exports created after the experiments; they do not define a new partition, were not retrospectively substituted into the pipeline, and do not alter any reported result.

Image Conditions

Every accepted synthetic molecule received a clean RDKit depiction. The matched augmented training condition applied controlled image transformations intended to approximate drawing variation while preserving the molecular label. Transformations included geometry, stroke, blur, contrast, noise, and background changes. Validation images remained clean for both conditions, so the synthetic comparison measured training-condition effects on the same data.

Held-Out Benchmark

The DECIMER Hand-Drawn Molecule Images v1.2 dataset contains 5,088 depictions drawn by 23 volunteers and is released under the Creative Commons Attribution 4.0 International licence (CC-BY 4.0) [3]. Archive metadata and published MD5 values were verified before extraction. The macOS metadata directory inside the archive was excluded, preventing the 5,088 actual PNG images from being double counted. The final manifest contained one unique label and image per row. Its SHA-256 digest is recorded in Appendix .

Model and Training Methodology

MolScribe Configuration

Table 3: Final MolScribe fine-tuning configuration

Component	Project setting
Encoder	Swin Base image encoder
Decoder	Six-layer Transformer decoder
Input	384×384 pixels
Output format	<code>chartok_coords</code> with 64 coordinate bins and separate x and y tokens
Starting weights	Official MolScribe <code>swin_base_char_aux_1m680k</code> checkpoint
Epochs	One matched epoch per condition
Batching	Batch size 2 with four-step gradient accumulation
Learning rates	Encoder 1×10^{-5} ; decoder 5×10^{-5}
Precision and device	Mixed precision on one Tesla T4
Random seed	20260915

The training run loaded the same official checkpoint for each condition. The encoder and decoder were fine-tuned jointly. Best-checkpoint saving used the shared clean validation set. Because the dataset supplied image-SMILES pairs but no atom-bond edge annotations, the project did not train MolScribe’s optional edge decoder. The custom demonstration loader therefore uses the trained sequence-and-coordinate output configuration.

Prediction Processing

MolScribe generated a raw token sequence for each image. Its post-processing converted valid outputs into SMILES suitable for evaluation. RDKit then parsed and canonicalized both the prediction and reference. A prediction counted as valid only if RDKit produced a sanitized molecule. Invalid predictions received zero exact-match and zero similarity scores and were never removed from the denominator.

Evaluation Metrics

- **Valid SMILES rate:** fraction of predictions parsed and sanitized by RDKit.
- **Exact isomeric match:** equality after canonicalization with stereochemistry retained.
- **Exact non-isomeric match:** equality after canonicalization with stereochemistry ignored.
- **Mean Tanimoto similarity:** Tanimoto coefficient between 2,048-bit radius-two Morgan fingerprints; invalid predictions score zero [5].

- **Paired 95% confidence interval:** percentile bootstrap of the per-image augmented-minus-clean difference using 2,000 resamples and seed 20260915.

The Tanimoto values are not directly interchangeable with the primary paper’s values because that paper used PubChem fingerprints implemented in the Chemistry Development Kit, whereas this project used RDKit Morgan fingerprints. Exact-match rates are the clearer cross-study comparison, although training scale and model architecture still differ substantially.

Development Experiments

The initial plan used a compact locally controlled convolutional encoder and Transformer decoder. These early experiments validated the pipeline and established the need for transfer learning. On all 470 clean validation images, the strongest scratch model generated 78 valid SMILES, no exact matches, and mean Tanimoto 0.0120. A tiny training-only probe memorized eight short molecules exactly, confirming that the training and decoding code paths operated correctly before the transfer-learning stage.

Table 4: Development experiments and modelling decisions

Development experiment	Main observation	Decision
Compact scratch encoder-decoder	78 of 470 valid and zero exact matches	Transfer learning selected
Short-molecule curriculum	1 of 470 valid; outputs were frequently too long	Not carried forward
Fivefold end-token weighting	68 of 470 valid and zero exact matches	Not carried forward
Native 224-pixel input	2 of 470 valid and 374 outputs over 100 characters	Not carried forward
Group normalization	Stable loss but zero valid outputs	Not carried forward
ImageNet MobileNet with fixed encoder weights	54 of 470 valid and zero exact matches	Transfer learning selected
Beam width five	One valid candidate among 320 sampled candidates	Greedy decoding retained
MolScribe transfer	Synthetic viability gate passed	Selected for matched study

These diagnostics changed the modelling route before the real benchmark was opened. They did not tune the final model to real images. The selected transfer approach balanced feasibility, open-source availability, and clear evidence that the model could decode chemical structures.

Synthetic Validation Results

Untouched Pretrained Model and Clean Fine-Tuning

Table 5: Untouched pretrained model versus clean fine-tuning on 470 synthetic validation images

Metric	Pretrained	Clean fine-tuned	Change
Valid SMILES	93.19%	81.28%	−11.91 pp
Exact isomeric	64.04%	63.83%	−0.21 pp
Exact non-isomeric	84.89%	78.94%	−5.96 pp
Mean Tanimoto	0.9057	0.8054	−0.1003

The official pretrained model substantially outperformed the clean-fine-tuned checkpoint on the clean synthetic validation set. This result does not prove independent superiority because overlap between MolScribe pretraining molecules and the ChEMBL-derived validation molecules cannot be ruled out. It does show that one epoch of narrow project fine-tuning can degrade a strong pretrained model.

Matched Clean and Augmented Fine-Tuning

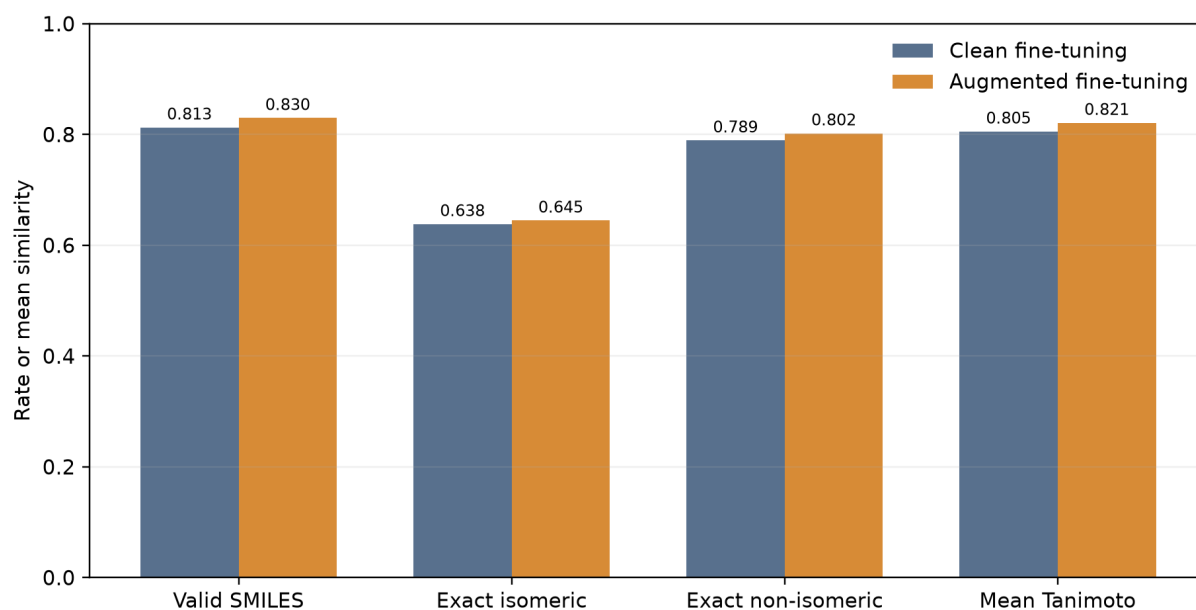


Figure 2: Matched results on 470 clean synthetic validation images.

Table 6: Matched comparison on clean synthetic validation

Metric	Clean	Augmented	Difference	Paired 95% interval
Valid SMILES	81.28%	82.98%	+1.70 pp	+0.21 to +3.40 pp
Exact isomeric	63.83%	64.47%	+0.64 pp	−1.06 to +2.55 pp
Exact non-isomeric	78.94%	80.21%	+1.28 pp	−0.43 to +2.98 pp
Mean Tanimoto	0.8054	0.8210	+0.0156	+0.0002 to +0.0316

Augmented training produced small improvements in validity and mean similarity on clean synthetic validation. Exact-match intervals included zero. These results justified evaluating the already selected pair on the real benchmark; they did not justify selecting or modifying an augmentation after observing real-image performance.

Held-Out Real Benchmark Results

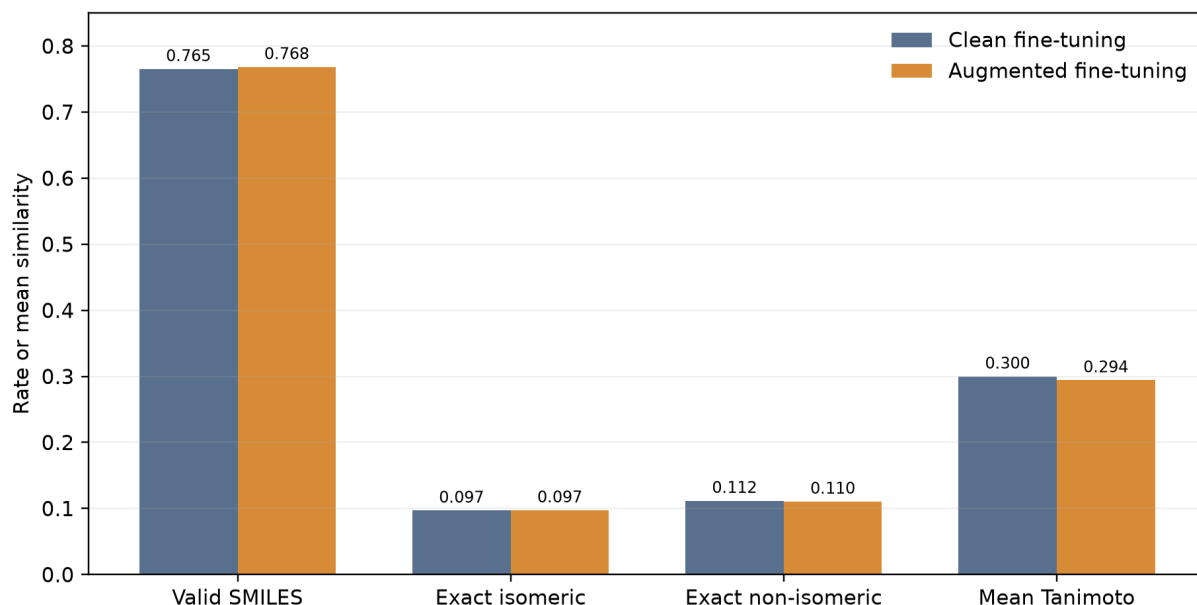


Figure 3: Performance on 5,088 held-out real hand-drawn images.

Table 7: Clean and augmented fine-tuned models on the held-out DECIMER benchmark

Metric	Clean	Augmented	Difference	Paired 95% interval
Valid SMILES	76.51% (3,893)	76.83% (3,909)	+0.31 pp	−0.69 to +1.28 pp
Exact isomeric	9.69% (493)	9.73% (495)	+0.04 pp	−0.47 to +0.53 pp
Exact non-isomeric	11.16% (568)	11.05% (562)	−0.12 pp	−0.65 to +0.39 pp
Mean Tanimoto	0.2998	0.2941	−0.0057	−0.0111 to −0.0005

The clean model’s mean-Tanimoto 95% interval was 0.2912 to 0.3087; the augmented model’s interval was 0.2852 to 0.3029. Median similarity was 0.2105 for clean and 0.2000 for augmented fine-tuning. Predictions at or above 0.8 similarity accounted for 11.50% and 11.38%, respectively.

Primary Finding

The real benchmark does not support the primary hypothesis. Validity and exact-match differences were near zero, and their paired intervals included zero. Mean Tanimoto similarity was slightly lower after augmented fine-tuning, with a paired interval that did not include zero. For this specific augmentation, checkpoint, data size, and one-epoch transfer design, the measured result is no improvement over clean fine-tuning.

Pairwise Outcome Changes

For exact isomeric recognition, 404 images were correct under both checkpoints and 4,504 were incorrect under both. Augmentation corrected 91 previously incorrect cases but lost 89 previously correct cases. Per-image Tanimoto improved for 960 images, tied for 3,078, and was lower for 1,050. This bidirectional movement explains why a few individual examples cannot establish a general improvement.

External High Accuracy Reference

The official DECIMER hand-drawn checkpoint was evaluated without modification as an external pre-trained reference. A fixed seed-20260925 simple random sample contained 100 benchmark images: 20 initial random images and 80 additional images selected from the remaining set. This sample was used only for measurement; no model setting was selected or changed from its results. Exact isomeric accuracy was 64.00%, with a Wilson 95% interval from 54.24% to 72.73%.

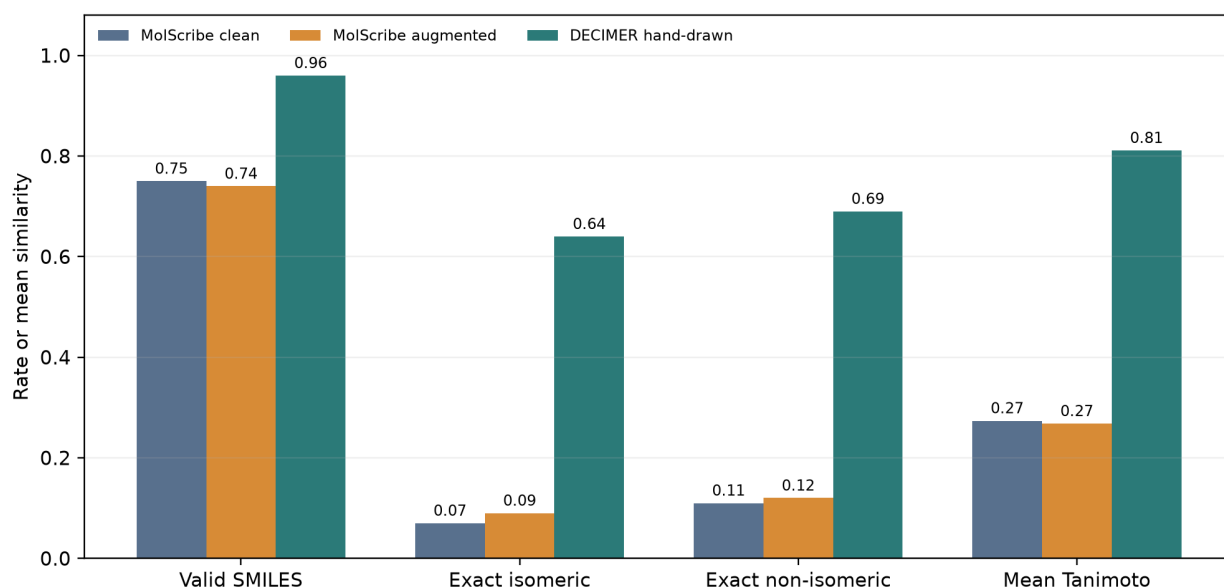


Figure 4: Fixed external reference on the same 100 benchmark images. DECIMER is an external pretrained model.

Table 8: Matched comparison on the fixed 100-image sample

Metric	MolScribe clean	MolScribe augmented	DECIMER hand-drawn
Valid SMILES	75.00%	74.00%	96.00%
Exact isomeric	7.00%	9.00%	64.00%
Exact non-isomeric	11.00%	12.00%	69.00%
Mean Tanimoto	0.2737	0.2681	0.8111

On the matched sample, DECIMER exceeded augmented MolScribe by 55 percentage points in exact isomeric accuracy; the paired bootstrap 95% interval was 45 to 65 points. Mean Tanimoto increased by 0.5431, with a paired interval from 0.4604 to 0.6241. These values support DECIMER as the demonstration model, but they do not replace the MolScribe fine-tuning experiment: DECIMER was trained externally at far greater scale and the project did not modify its weights.

Subgroup and Error Analysis

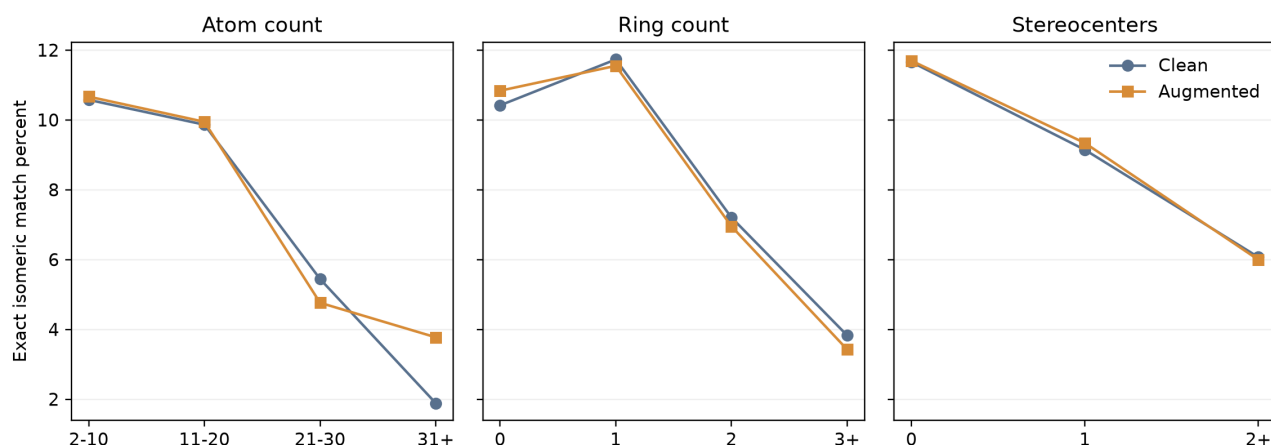


Figure 5: Exact isomeric accuracy by molecular complexity.

Molecular complexity was more informative than the training condition. For the clean model, exact isomeric accuracy declined from 10.57% for structures with 2 to 10 atoms to 5.44% for 21 to 30 atoms. It declined from 10.42% for acyclic molecules to 3.83% for molecules with at least three rings. References with two or more stereocenters reached 6.07% exact accuracy, compared with 11.66% when no stereocenter was present.

Most subgroup augmentation intervals included zero. The clearest exception was the 26 to 50 character reference-SMILES group, where mean Tanimoto changed from 0.3238 to 0.3145. The difference was -0.0092 with a paired 95% interval from -0.0173 to -0.0013 . The 31-plus atom and 76-plus character groups were small and had wide intervals, so their apparent gains must not be treated as evidence of benefit.

Observed Failure Modes

- Invalid or truncated SMILES sequences, especially for longer structures.
- Incorrect ring closure or branch connectivity that preserved fragments but changed the graph.
- Loss or inversion of stereochemistry, reflected by lower accuracy with multiple stereocenters.
- Atom-label and charge errors in visually crowded regions.
- Reduced recognition for ring-rich and larger molecules where local line errors affect more of the structure.

A high valid-SMILES rate did not imply correct recognition. Both models produced valid outputs for more than three quarters of the images but exactly matched fewer than one in ten when stereochemistry was retained. Chemical validity therefore measures grammatical plausibility, not agreement with the drawing.

Discussion

Why Synthetic Gains Did Not Transfer

The augmented model improved slightly on clean synthetic validation but not on real drawings. The most likely explanation is a remaining gap between the synthetic perturbations and the real drawing

distribution. Simple geometric and image perturbations can imitate blur, rotation, noise, or stroke variation, but they do not fully reproduce how people simplify rings, position labels, draw stereochemical wedges, or create ambiguous intersections. One epoch on 3,758 molecules may also be too small to alter a large pretrained model reliably without eroding useful prior representations.

Comparison with the Primary Reference

Rajan et al. reported 73.25% identical predictions on the DECIMER hand-drawn benchmark, while the present project achieved about 9.7% exact isomeric recognition. The gap is expected: the reference trained for 25 epochs on TPU V4 hardware using millions to tens of millions of molecules and multiple synthetic hand-drawn depictions per molecule. This project fine-tuned for one epoch on 3,758 molecules. The architectures, tokenization, fingerprint metric, and training supervision also differ. The comparison shows the cost of scale and domain-specific training; it is not a like-for-like leaderboard result.

Meaning for the Research Questions

Table 9: Answers to the research questions

Question	Answer supported by this study
RQ1	The clean project checkpoint achieved 9.69% exact isomeric accuracy and 0.2998 mean Tanimoto.
RQ2	The tested augmentation did not improve the real benchmark and slightly reduced mean similarity.
RQ3	Larger, ring-rich, and stereochemically complex structures were harder; small subgroups remained uncertain.
RQ4	Validity suggested plausible syntax, while exact and similarity metrics exposed large structural errors.

Technical Considerations and Study Scope

Implementation Notes

- The original compact architecture established pipeline viability; the final study used an OCSR-specific transfer model for stronger decoding performance.
- MolScribe required an isolated Python 3.10 environment and older pinned Torch, NumPy, OpenNMT, and vision-library versions.
- The DECIMER image archive contained Apple metadata PNG entries, which initially doubled the apparent image count until path filtering was corrected.
- The project checkpoints contain only `chartok_coords` supervision; the stock MolScribe single-image interface expects an edge decoder and required a compatible custom loader.
- CPU inference for the external DECIMER reference averaged 4.32 seconds per image, so the project used a fixed 100-image sample and reported its uncertainty rather than presenting a partial full-benchmark run.

Study Scope and Interpretation Boundaries

- Training used 3,758 molecules and one fine-tuning epoch; comparisons with the reference study therefore reflect different training scales.

- The augmented condition combined several perturbations, so this completed experiment cannot identify which transformation helped or harmed.
- Synthetic validation used clean images; it did not independently validate robustness to synthetic drawing styles.
- Potential molecule overlap with MolScribe pretraining data was not assessed, so the untouched pretrained synthetic result is interpreted separately from project fine-tuning.
- Two benchmark references were not parsed by the local RDKit version and received zero chemical scores, although both stayed in the denominator.
- The project reports model recognition, not suitability for unsupervised scientific or clinical use. Every predicted structure requires verification.

Reproducibility and Assessment

The repository separates source code, manifests, configurations, saved predictions, and metric summaries. The fastest assessment path recomputes every reported benchmark metric directly from the two saved per-image prediction files. It verifies file hashes, row count, ID order, summary values, paired differences, confidence intervals, and transition counts. This path does not require the 1.1 GB checkpoints, Colab, or a GPU.

Table 10: Assessment pathways

Assessment level	Required resources	Purpose
Fast verification	Local Python environment and saved CSV files	Recompute the saved benchmark metrics
Prediction audit	Saved native and per-image outputs	Inspect any benchmark row and metric input
Full inference	T4 GPU, DECIMER images, project checkpoint	Regenerate predictions; optional for assessment
Full training	T4 GPU, synthetic package, official starting checkpoint	Repeat one matched fine-tuning condition

General setup and verification commands

```
python -m venv .venv
# Activate .venv using the command for the operating system.
python -m pip install -r requirements.txt
python -m pip install -e . --no-deps
python scripts/reproduce_benchmark_metrics.py
python -m pytest -q
python scripts/compare_decimer_external_sample.py
python scripts/run_saved_prediction_demo.py
```

At the time of final report preparation, 36 automated tests passed. The tests cover chemistry normalization, metrics, splitting, image handling, tokenizer behavior, model components, comparison logic, Colab-package integrity, prediction import, and demonstration checkpoint verification.

The external DECIMER sample is preserved as 100 per-image prediction rows together with a machine-readable summary and a matched comparison against both MolScribe checkpoints. The sample definition, output hashes, paired differences, and intervals can therefore be audited without rerunning TensorFlow inference.

Conclusions and Evidence-Based Next Steps

This project built a complete and auditable image-to-SMILES study using course-project resources. It established a viable MolScribe transfer pipeline, trained matched clean and augmented models, selected model files using clean validation data before testing, and evaluated every image in a real hand-drawn benchmark. The main result is that the tested augmentation did not improve exact recognition and produced a small reduction in mean Morgan-fingerprint similarity. Keeping the held-out result unchanged by post-test tuning preserves the validity of the comparison.

For a high-accuracy demonstration, the unchanged official DECIMER hand-drawn model is the stronger choice. It achieved 64.00% exact isomeric accuracy and 0.8111 mean Tanimoto on the fixed 100-image sample, compared with 9.00% and 0.2681 for augmented MolScribe on the same images. This external result shows the benefit of large-scale hand-drawn-specific training; it is reported separately and is not presented as a project-trained checkpoint.

Evidence from the subgroup and error analyses supports increasing chemically diverse training data, using a depiction generator that models human conventions more directly, retaining full atom–bond graph supervision, and evaluating augmentation families separately before a newly held-out test. Longer transfer schedules should be selected with synthetic validation and an independent development set, not the current benchmark. Confidence calibration or human-review triage can also support assisted curation. These changes should be evaluated on a newly held-out set rather than by repeatedly tuning against the already opened 5,088-image set.

Acknowledgement of AI Assistance

OpenAI Codex and ChatGPT were used to assist with code drafting and debugging, workflow organization, documentation structure, and language editing. The author reviewed and executed the computational work, checked the saved outputs, and retained responsibility for the research questions, experimental controls, interpretation, and final scientific claims. AI-generated suggestions were not treated as evidence; all quantitative results reported in this thesis are derived from the preserved project code, prediction files, and reproducibility records.

References

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- [2] Y. Qian, J. Guo, Z. Tu, Z. Li, C. W. Coley, and R. Barzilay, "MolScribe: Robust molecular structure recognition with image-to-graph generation," *Journal of Chemical Information and Modeling*, 2023. <https://doi.org/10.1021/acs.jcim.2c01480>
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- [7] D. Mendez, A. Gaulton, A. P. Bento, et al., "ChEMBL: towards direct deposition of bioassay data," *Nucleic Acids Research*, vol. 47, no. D1, pp. D930–D940, 2019. <https://doi.org/10.1093/nar/gky1075>
- [8] RDKit contributors, "RDKit: Open-source cheminformatics software." <https://www.rdkit.org/>

Appendix A: Reproducibility Record

Table 11: Reproducibility identifiers and verification status

Artifact	Recorded value
MolScribe source commit	7296a30413eb55436702011efdff78131f66d162; recorded in both run-information files
Official starting checkpoint SHA-256	32821fa4ba8d17f9a92e374601438c5e757947dbf1141e9ad0ef51cccfa1c501; recorded in both Colab run-information files; binary stored externally
Clean final checkpoint SHA-256	64a5bf5c356cd535cf0fdc76ef26164bfe00d3ab837c0376d2334ca7e663e3fd; recorded from the clean Colab output; binary linked from the README
Augmented final checkpoint SHA-256	20bba8c832365f1b708cfe7738b5dfd5f90b025bf6a4619288240e0c6f313741; recorded from the augmented Colab output; binary linked from the README
Clean package SHA-256	05c42002ea761ce5ff265948cdabffef3329a5ca3b63007a758d2a4d66164252; recomputed from the 27,348,476-byte archive
Augmented package SHA-256	52ccd5c5eac61cd01ca55db36752da8c729ab322b628ff09b14af8bd2e90356c; recomputed from the 123,810,243-byte archive
DECIMER benchmark manifest SHA-256	bb88840799a54eadbaff7776a440744dec2cdeb7a84bde37d8cb4294e58e97f8; recomputed from the 5,088-row benchmark CSV
DECIMER 100-image prediction CSV SHA-256	444e094eccd1c41dfba3a98daa4871ecd1c220bc285687933803a0a2a7aed58; recomputed from the submitted CSV
DECIMER 100-image summary JSON SHA-256	26c34cc2e03c1a21d70558de3e7967a97a25af433cc45abbd13aada76a65ad40; recomputed from the submitted JSON
Training hardware	Google Colab Tesla T4
Isolated Python	3.10.21
Core framework	PyTorch 1.13.1 with CUDA 11.7
External reference package	DECIMER 2.8.0
External reference runtime	Python 3.13.15, TensorFlow 2.20.0, RDKit 2026.03.6
External reference device	Google Colab CPU

Appendix B: Submitted Repository Map

Table 12: Repository contents supplied with the project

Location	Contents
README.md	Project summary, general execution instructions, checkpoint links, and interpretation notes
requirements.txt, pyproject.toml	Pinned verification dependencies and installable package metadata
src/ocsr_project	Chemistry, dataset, model, training, and evaluation modules
scripts	Runnable data preparation, training, import, analysis, and verification tools
configs	Versioned experiment configurations
data/raw, data/manifests	ChEMBL source tables, provenance metadata, and exact scaffold-separated split assignments
splits	Standalone train, validation, and test CSV files containing the exact submitted assignments
data/processed	Preserved validation image used by the saved-prediction demonstration
artifacts/metrics	Saved predictions, summaries, comparisons, confidence intervals, and subgroup outputs
artifacts/demo	Generated saved-prediction demonstration panel
reports/final	Thesis PDF and source, presentation PDF and source, and report figures
colab	Pinned MolScribe training and single-image demonstration notebooks
demo	Evaluator-facing Gradio interface for a new single-image input
tests	Automated unit and integration tests